Skills Summary

Factor Instead of Dividing, and Check Every Candidate

Use this reference while completing your worksheet or when studying.

Skill 1 — Factor, never divide

Zero-product property: if $PQ = 0$ then $P = 0$ *or* $Q = 0$ — both branches, always.

- Dividing by an expression assumes it is not zero, which quietly deletes every solution that makes it zero.
- Move everything to one side, factor, then solve each branch separately.
- The lost solutions are exactly the zeros of whatever was divided by — useful for diagnosing someone else's work.

Example: Solve $2\sin\theta\cos\theta = -\sin\theta$ on $0^\circ \le \theta < 360^\circ$.
Step 1. Both sides share a $\sin\theta$, which is the temptation to divide, so move everything to one side and factor instead to keep that branch alive.
Step 2. $\sin\theta\,(2\cos\theta + 1) = 0$, so $\sin\theta = 0$ or $\cos\theta = -\frac{1}{2}$
The solutions are $\boldsymbol{\theta = 0^\circ,\ 120^\circ,\ 180^\circ,\ 240^\circ}$.

Try it: Solve $2\sin\theta\cos\theta = \cos\theta$ on $0^\circ \le \theta < 360^\circ$.

check: $\boldsymbol{\theta = 30^\circ,\ 90^\circ,\ 150^\circ,\ 270^\circ}$

Skill 2 — Only factors cancel

The rule: $\dfrac{ab}{a} = b$, but $\dfrac{a+b}{a} \neq b$.

- A denominator divides the *whole* numerator, so it cancels only against a factor common to every term.
- Factor first (difference of squares, common factor), then cancel; or split the fraction term by term, which is always legal.
- One numerical test kills a wrong simplification: pick a convenient value and compare the two forms.

Example: Is $\dfrac{\cos^2 x - 1}{\cos x - 1}$ equal to $\cos x$? Test it where $\cos x = 0.6$.
Step 1. A suspected illegal cancellation is fastest to settle numerically, so evaluate the original expression at one convenient value and compare.
Step 2. $\dfrac{0.36 - 1}{0.6 - 1} = \dfrac{-0.64}{-0.4}$, while $\cos x$ would give 0.6
The true value is **1.6**, so the cancellation is wrong; factoring gives $\cos x + 1$.

Try it: Evaluate $\dfrac{\cos^2 x - 1}{\cos x - 1}$ where $\cos x = 0.25$.

check: **1.25**

Skill 3 — Keep both signs, check every candidate

Square roots: $\sqrt{u^2} = |u|$, so $u^2 = k$ gives $u = \pm\sqrt{k}$ — two branches. **Squaring both sides** can add solutions: $A^2 = B^2$ only forces $A = \pm B$.

- Safer than rooting: factor as a difference of squares, which shows both branches without any sign to remember.
- Squaring *adds* solutions; dividing *removes* them. Only the first can be repaired afterwards — by testing every candidate in the original equation.

Example: Solve $4\cos^2\theta = 1$ on $0^\circ \le \theta < 360^\circ$.
Step 1. Rooting both sides risks dropping the negative branch, so factor the difference of squares and let each factor supply its own sign.
Step 2. $(2\cos\theta - 1)(2\cos\theta + 1) = 0$, so $\cos\theta = \pm\frac{1}{2}$
The solutions are $\boldsymbol{\theta = 60^\circ,\ 120^\circ,\ 240^\circ,\ 300^\circ}$.

Try it: Solve $4\sin^2\theta = 2$ on $0^\circ \le \theta < 360^\circ$.

check: $\boldsymbol{\theta = 45^\circ,\ 135^\circ,\ 225^\circ,\ 315^\circ}$

(!) Watch out: If your solution set is a strict subset of a classmate's and both of you did correct arithmetic, look for a division — the missing angles are the zeros of whatever you divided by.